AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 21 · STUDENT LESSON

Good Data, Bad Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Check the source before changing the value.

Audit schemas and units, preserve measured zero and missingness, and clean a new flight log with traceable evidence.

Keep one habit: Keep the raw record → justify each change → report assessed and unresolved records.

MAIN LESSON · SECTION 1

Which changes does the evidence support?

> Check schema, units, identity and source records without turning unknown into zero.

Start with a tidy-looking table

A fully filled table can contain invented values, while an honest table still has missing observations. Keep the raw records and link each correction to evidence before counting results.

Retrieve the missing-availability limit

A recommender needs availability, but that field is blank. Can the highest preference score establish that the item is available? Explain.

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can check records against a stated format, unit and meaning before using them.
- I can distinguish an **evidence**-based **correction** from an unsupported guess.
- I can keep original values and a **change log**, then report what remains unresolved.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confidence Is Not **Correctness**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

Coverage, **correctness**, and **relevance** matter more than raw quantity.
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LEARNING TARGET A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can repair data with traceable **evidence** while preserving valid zero and unknown.

MAIN LESSON · SECTION 2

Read the schema and purpose before cleaning

> RampLog is a separate fictional dataset with centimetres for both height and travel.

Separate schema validity from truth

A tidy table can still contain wrong information. **Validation** checks whether a record follows a **schema**. A **schema** says which fields are required and what their values mean.

Passing these checks makes a record usable for a stated task; it does not prove that the measurement or claim is true.

Read the required fields and ranges

Our fictional RampLog records one ramp trial per trial ID. The fields are trial ID, height in centimetres, travel in centimetres and source card. Height must be a whole number from 1 through 10.

Travel must be a whole number from 0 through 10. The source card identifies the original **observation**. Repeated trial IDs need investigation before counting trials.

State the counting task and zero/missing distinction

The purpose is to count trials whose travel is at least 4 cm. A measured zero means no travel; a blank means no value was recorded. Never silently replace a blank by zero.

Use only the invented records here; no personal information is needed.

Check what each field means

- Which RampLog fields have different allowed ranges? Why can travel zero be a valid measurement while a blank travel field cannot be counted as zero? What identifies one distinct trial?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY One animal dataset has 1,000 blurry **duplicate** photos; another has 120 clear, correctly labeled photos across settings and animal types.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Correct a copy while preserving the raw record

> Use source cards to support a value change and confirm whether repeated rows are one trial.

Compare three evidence situations

Preserve the original and explain each decision

ORIGINAL RECORD	EVIDENCE CHECK	WORKING DECISION
T2 height 60 cm	Card C2 says 6 cm	Correct to 6; log C2
T3 travel blank	Card C3 also blank	Keep missing; flag
T4 travel 0 cm	Card C4 confirms zero	Keep measured zero

C2 supports correcting T2. C3 cannot supply a missing measurement. C4 shows that zero is a recorded observation, not a placeholder for missing data.

Read the complete raw and source records

The complete raw table is T1: height 4, travel 3, card C1; T2: height 60, travel 5, card C2; T3: height 5, travel blank, card C3; T4: height 6, travel 0, card C4. A second export row repeats T1 with height 4, travel 3 and C1.

C1 confirms one trial T1, not two. C2 reads height 6 and travel 5. C3 has no travel measurement. C4 reads height 6 and travel 0.

Write a traceable correction and duplicate link

Work in a separate corrected copy. Record T2 height 60 to 6, **evidence** C2, transcription error, reviewer and review date. Link the repeated T1 export row to the first T1; count that single trial once and keep both raw rows.

Do not remove two different trials merely because their measured numbers match.

Explain one supported change

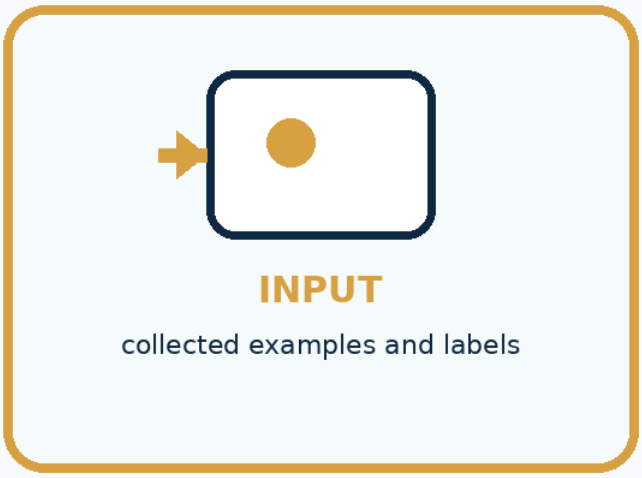
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

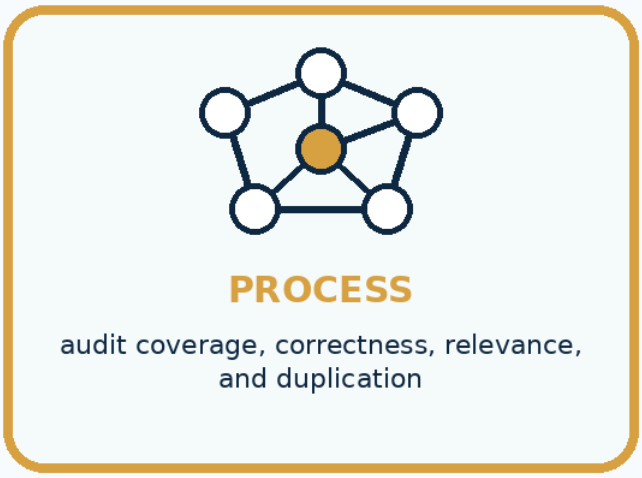
▼ Earlier wording, alternatives and picture transcripts

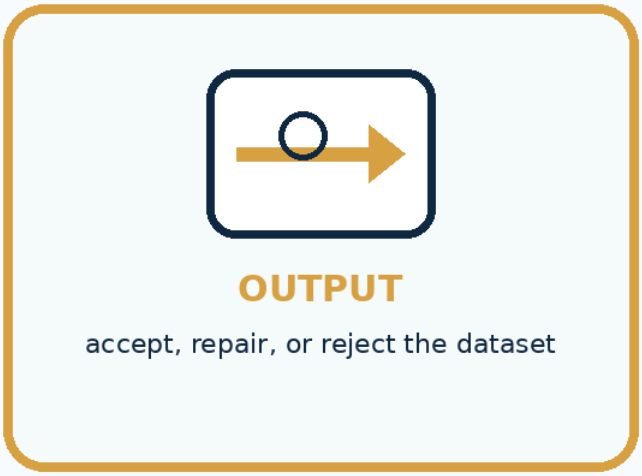
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT collected examples and labels

PROCESS audit **coverage**, **correctness**, **relevance**, and duplication

OUTPUT accept, repair, or reject the dataset

CORE IDEA **Coverage**, **correctness**, and **relevance** matter more than raw quantity.

UNDER THE HOOD Data quality is multidimensional. More rows can amplify **duplicate**s, systematic labeling errors, stale conditions, or irrelevant fields.

Fitness depends on the intended population and decision, not a universal quality score.

Draw a fresh example with all three stages:

A closer explanation

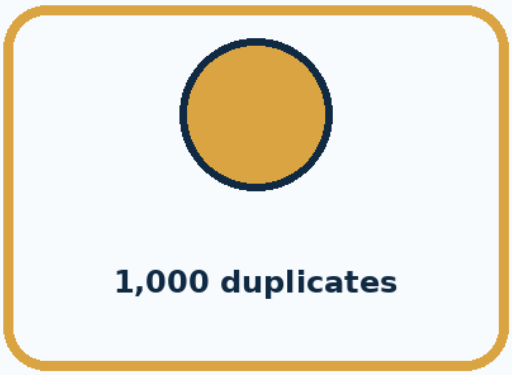
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare quality problems and evidence-backed actions

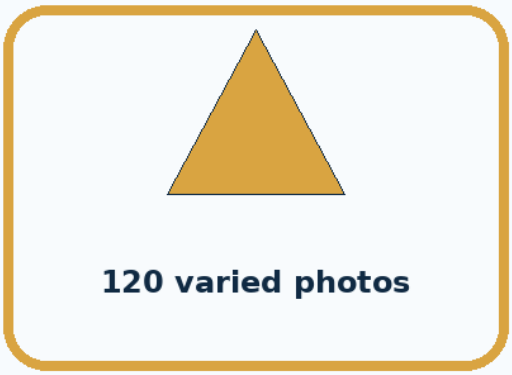
> Distinguish known units, missing observations, conflicting sources and repeated copies.

Pictures paired with their evidence



The case to investigate

CASE 1 — 1,000 **duplicate**s Clue: high count, low variety. Meaning: weak **coverage**.



The case to investigate

CASE 2 — 120 varied photos Clue: settings and types represented. Meaning: stronger **coverage**.



The case to investigate

CASE 3 — Wrong labels Clue: targets conflict with reality. Meaning: **correctness** failure.



The case to investigate

CASE 4 — Owner birthday Clue: unrelated to animal type. Meaning: irrelevant/private.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

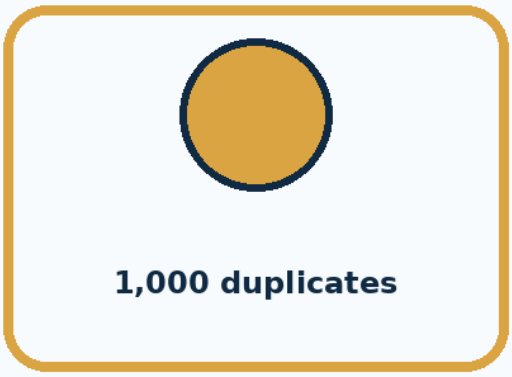
BOUNDARY Large quantity cannot compensate for systematic absence or incorrect labels.

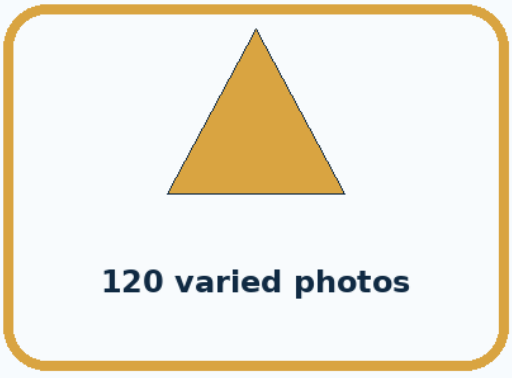
▼ Earlier wording, alternatives and picture transcripts

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — 1,000 **duplicate**s Clue: high count, low variety. Meaning: weak **coverage**.

CASE 2 — 120 varied photos Clue: settings and types represented. Meaning: stronger **coverage**.

CASE 3 — Wrong labels Clue: targets conflict with reality. Meaning: **correctness** failure.

CASE 4 — Owner birthday Clue: unrelated to animal type. Meaning: irrelevant/private.

Change one thing in the pictured case

Change: The source log gives B only as bare 200 with no unit.

Prediction: B remains unresolved rather than becoming 20 cm.

Reason: Without a confirmed unit, division by 10 would be an unsupported assumption.

MAIN LESSON · SECTION 6

Keep zero, missing and documented unit repair separate

> Use the toy-car contract and original source evidence before changing the exported copy.

The raw toy-car rows and verification policy

A fictional toy-car distance table needs cleaning before use.

Contract: unique run ID; whole distance 0–30 cm. Raw rows: A=0 cm, B=200 mm, C=blank, another A=0 cm.

Source log confirms one A run and B's original measurement of 200 mm; C has no measurement.

This toy-car activity audits export claims against the original source log. Confirm B's original unit from that source before authorising its conversion. A unit printed in the export is a claim being checked, not an independently confirmed source value.

Compare record states and supported repairs

AI ACADEMY / AI-1/21 · 03 · INSPECT THE LESSON MODEL

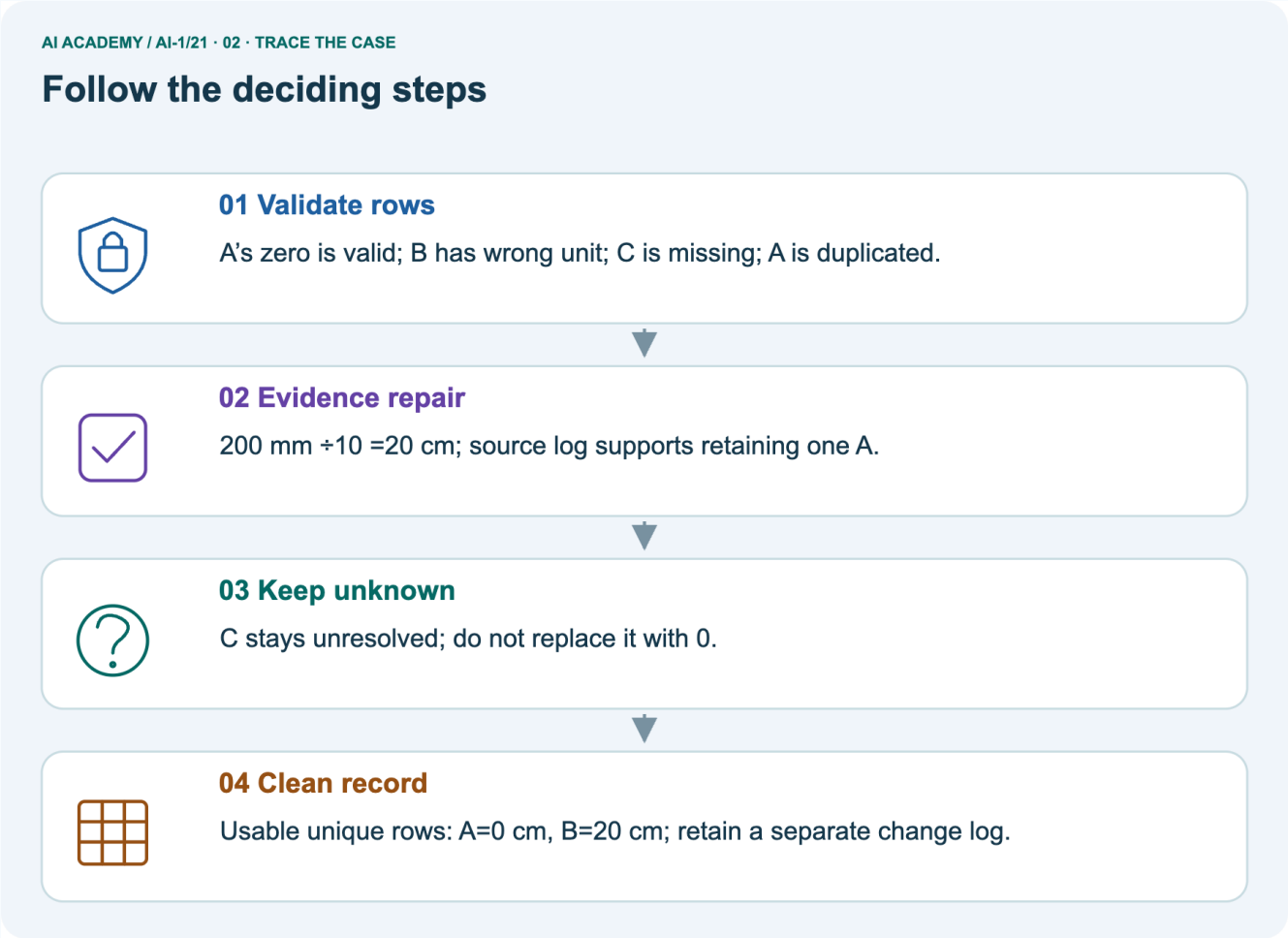
Preserve the original and explain each decision

Original record	Evidence check	Working decision
T2 height 60 cm	Card C2 says 6 cm	Correct to 6; log C2
T3 travel blank	Card C3 also blank	Keep missing; flag
T4 travel 0 cm	Card C4 confirms zero	Keep measured zero

C2 supports correcting T2. C3 cannot supply a missing measurement. C4 shows that zero is a recorded observation, not a placeholder for missing data.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace conversion, duplicate handling and retained unknown



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what authorises a correction

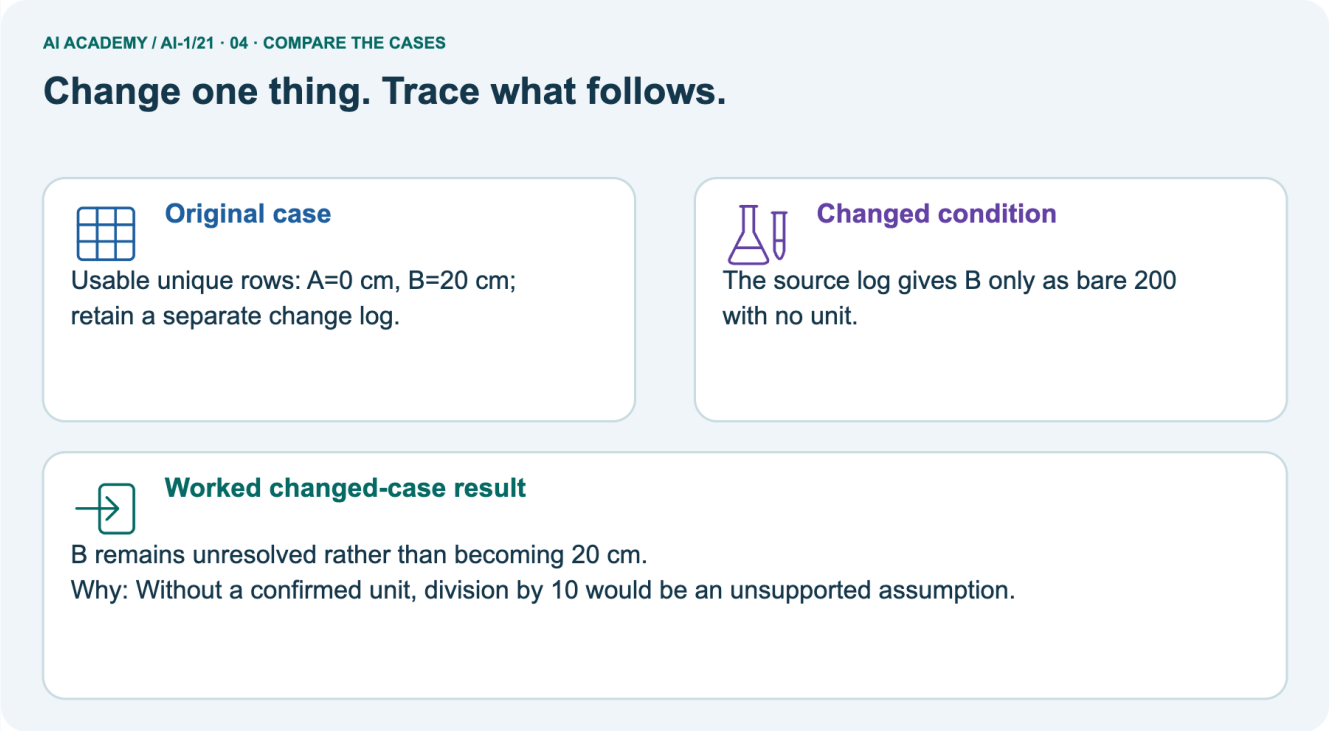
Cleaning is not permission to make inconvenient values plausible. The source log supports a unit conversion and removal of a **duplicate** copy. Zero means a measured zero distance; blank means no supplied measurement.

Keep the raw table and explain every change so someone can reconstruct what happened.

Predict when the source cannot establish the unit

- Before checking, suppose the original source can give only the digits 200, with no unit. Under this source-verification exercise, is B's original unit established well enough to authorise the conversion? State the unresolved evidence to request.

▼ Check the unresolved-source case



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit what clean formatting establishes

A clean table can still contain measurement or labelling errors that these checks do not detect.

Identify the supported original repairs

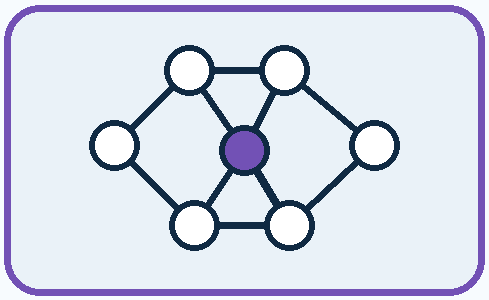
- Read the labelled picture: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?

▼ **Earlier wording, alternatives and picture transcripts**

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE One animal dataset has 1,000 blurry **duplicate** photos; another has 120 clear, correctly labeled photos across settings and animal types.

STEP 1 — OBSERVE collected examples and labels

STEP 2 — TRACE audit **coverage**, **correctness**, **relevance**, and duplication

STEP 3 — CONCLUDE accept, repair, or reject the dataset

MODEL REASONING Choose the set that covers intended conditions, uses checked labels, includes relevant fields, and avoids misleading duplication. State remaining gaps rather than declaring any dataset perfect.

DO NOT OVERCLAIM Large quantity cannot compensate for systematic absence or incorrect labels.

What new **evidence** could change this answer?

A closer explanation

Problem	Permitted action here	Unsupported shortcut
Invalid height 60	Check C2; correct to 6	Guess by deleting a zero
Missing travel	Keep missing; seek source	Fill with the class average
Repeated T1 ID	Confirm one trial on C1; count once	Treat every repeated number as a duplicate
Two sources disagree	Flag conflict; ask for resolution	Pick the value that improves the result

A conversion changes units while preserving the quantity. If an authorised source records 50 mm, that is 5 cm because 10 mm equals 1 cm. Write the original unit, conversion and resulting unit in the log.

A bare value 50 without a known unit does not justify assuming millimetres.

A plausible value can still be a transcription error: 7 would pass the height rule even if its source said 6. Conversely, an unusual but supported value is not automatically wrong.

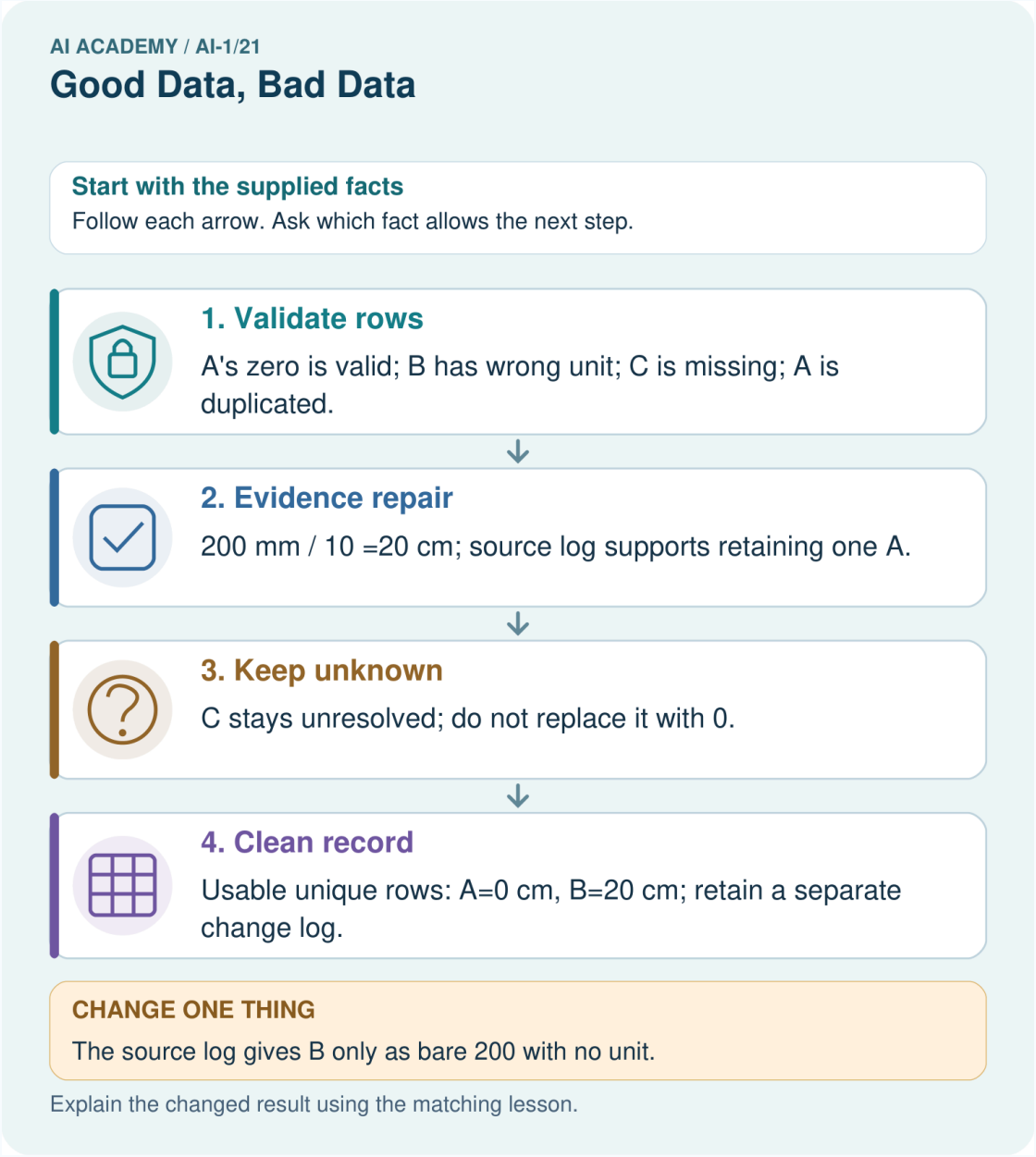
Checking format and checking agreement with a source are different jobs. A source can itself contain an error, so retain its identity and any unresolved concern.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Good Data, Bad Data

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate rows	A’s zero is valid; B has wrong unit; C is missing; A is duplicated.
2. Evidence repair	200 mm ÷10 =20 cm; source log supports retaining one A.
3. Keep unknown	C stays unresolved; do not replace it with 0.
4. Clean record	Usable unique rows: A=0 cm, B=20 cm; retain a separate change log.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: The source log gives B only as bare 200 with no unit. Predict the result before checking the explanation. What remains the same?

Recheck after repair and retain unresolved records

> A correction needs the right row, supported source and a separate account of what remains unknown.

Audit the corrected copy against its change log

After changing a value, run the **schema** checks again and compare the corrected copy with the raw copy and log. Recount distinct trials.

Check that every **correction** has **evidence** and that zero, missing and conflict remain separate. A repair can introduce a new mistake if a value is changed in the wrong row.

Limit the result to the covered records

The corrected RampLog supports a statement about its three assessed trials. It does not establish how every ramp or future trial behaves. Better formatting cannot create missing **observation**s or broaden what was sampled.

Next week asks which cases a dataset actually covers.

Repair the no-blanks-means-good-data claim

- CLAIM A The bigger dataset must be better.
- CLAIM B Clean formatting means the data is correct.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A table now has no blank cells because a designer guessed all missing measurements. Explain why this does not establish better evidence, and state what a trustworthy repair record should retain

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
1,000 duplicates	high count, low variety	weak coverage
120 varied photos	settings and types represented	stronger coverage
Wrong labels	targets conflict with reality	correctness failure
Owner birthday	unrelated to animal type	irrelevant/private

Repair two tempting claims

Repair:

Repair:

A closer explanation

Practise: count distinct assessed trials honestly

> Use verified travel values and show the missing trial separately.

Read the corrected RampLog summary

After the documented **correction** and confirmed **duplicate** handling, RampLog contains four distinct trials: T1, T2, T3 and T4. Three have recorded travel. T1 has 3 cm, T2 has 5 cm and T4 has 0 cm.

Only T2 meets at least 4 cm, so the result is 1 of 3 assessed trials, with 1 trial missing travel. Do not report T3 as a failure or hide it from the account.

> **A reviewer fills T3 travel with 4 because it makes the results look more balanced. Identify the unsupported change, write an honest report using the recorded evidence, and name the source information still needed**

Keep missing old observations distinct from new trials

Describe which records entered the calculation and why. Keep unresolved records visible in a separate status list.

A new measurement is a new **observation** with its own time and identity; it should not be passed off as the missing **observation** from an earlier trial.

Use W2 to write the supported correction

Use workbook W2 for the connected example “Data quality and documented repair”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “every valid value is true”

- Use the worked case for Data quality and documented repair. Proposed explanation: “Every valid value is true”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the toy-car source evidence

Which repairs are **evidence**-backed in the original case, and which row must remain unknown?

> **My explanation using a pictured step and the given facts**
>
> | |
> | --- |
> | |
> | |
> | |
> | |

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Coverage**, **correctness**, and **relevance** matter more than raw quantity.

Use a fresh example to explain **coverage**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Clean formatting means the data is correct.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Write a traceable correction

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate two mini-dataset repairs

> Use supplied raw and source cards to distinguish a documented conversion from an invented value.

The schema, task and original records

Audit two copies of fictional run records for counting distances at least 5 cm. Schema: unique run ID and whole distance 0–20 cm. Raw records: M1=4 cm, M2=blank, M3=0 cm, M4=80 mm. Original source cards confirm M1=4 cm, no measurement for M2, M3=0 cm and M4=80 mm. Keep these raw records unchanged.

Compare copies and calculate from supported evidence

1. Copy A lists 4,5,0,8 cm for M1–M4; it filled M2 with 5 without new evidence and has no change log. Copy B keeps M2 missing, preserves zero, and logs M4's conversion 80 mm÷10=8 cm with its source. Predict which copy supports the stated calculation better. Audit every change, report the assessed and missing counts, and write the source information still needed. Neither appearance nor a filled cell establishes a measurement.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep the audit trail and unresolved observation

Case	Prediction	Observed result	What it means
Normal			
Edge			

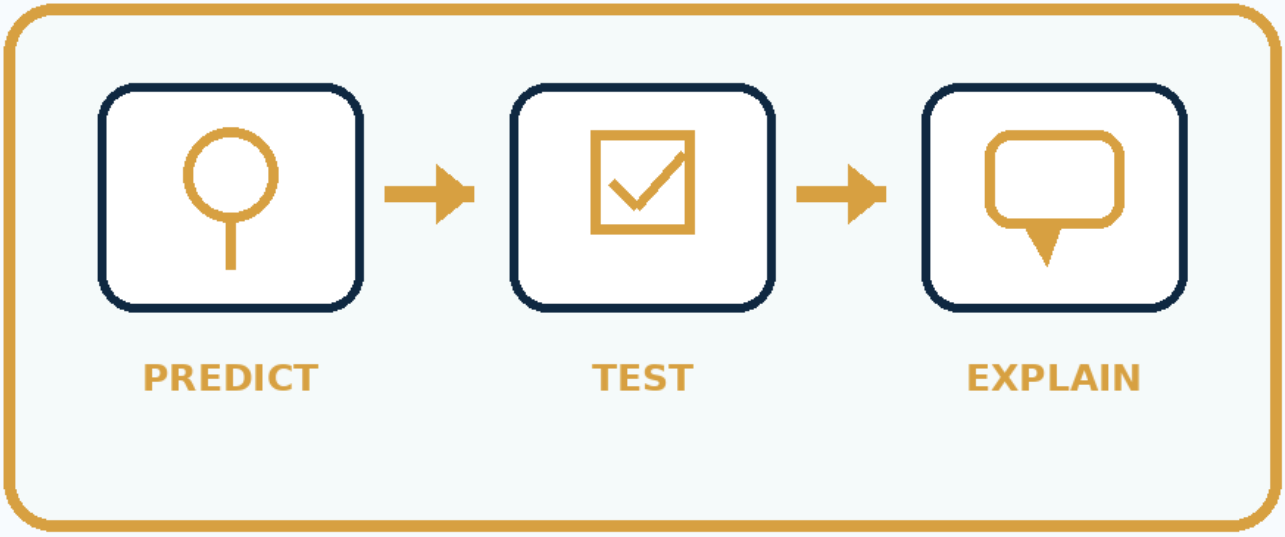
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EXPLORER LAB

Dataset Quality Audit

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Audit two mini datasets and choose the safer one.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the new paper-plane log

> Use the new flight IDs, source cards and distance range; retain the original rows.

The new schema and supplied evidence

Each flight ID must be unique; distance is a whole number from 0 through 20 cm. Raw rows: F1 8 cm; F2 120 cm; F3 blank; F4 0 cm; another F1 8 cm. Source S1 confirms one F1 flight. S2 says F2 travelled 12 cm. S3 is blank.

S4 confirms zero. Preserve the raw rows, describe supported repairs and give the number of distinct flights. Report how many assessed flights travelled at least 10 cm, including missing records separately.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Audit an independent paper-plane log

Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A table labels a cat photo dog and lists the same photo three times. Identify both data problems and one sensible repair for each.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Audit the sorter's example cards and add a reason for including each field and condition.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep raw, validated and corrected sorter records distinct

> Link every repair to the evidence and preserve the reason for unresolved fields.

This week’s project decision

THIS WEEK'S CHECKPOINT Audit the sorter's example cards and add a reason for including each field and condition.

Keep raw sorter **input**s, validated records and an **evidence**-linked repair log as different artifacts.

Plan schema checks, sources and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Keep coverage and correctness visible alongside quantity

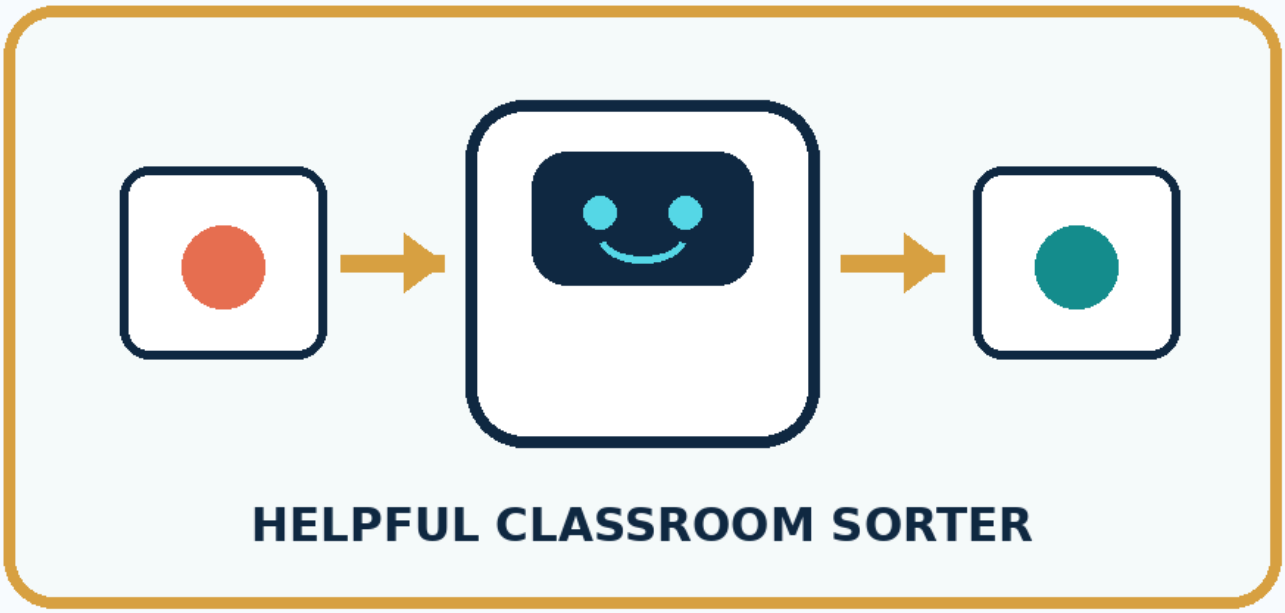
RESPONSIBLE DESIGN Large quantity cannot compensate for systematic absence or incorrect labels.

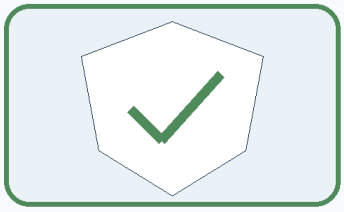
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain why each change is supported

> Close your notes and distinguish a conversion, correction, duplicate link and unresolved observation.

Your exit explanation

- Close your notes. Explain this goal in your own words: Check records against a stated format, unit and meaning before using them

My deciding step, evidence and one remaining question

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
coverage	how well examples represent needed situations	
correctness	whether records and labels are accurate	
relevance	whether data supports the task	
duplicate	repeated record adding little new coverage	

Terms for the connected explanation

Word	Meaning
Schema	A description of required fields, formats, units and allowed values
Validation	Checking a record against stated requirements
Provenance	A record of where information came from

Word	Meaning
Correction	A change supported by evidence about the intended value
Change log	A record of what changed, why, when and on what evidence

Repair a claim using a word

- CLAIM TO REPAIR The bigger dataset must be better.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ **Earlier wording, alternatives and picture transcripts**

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

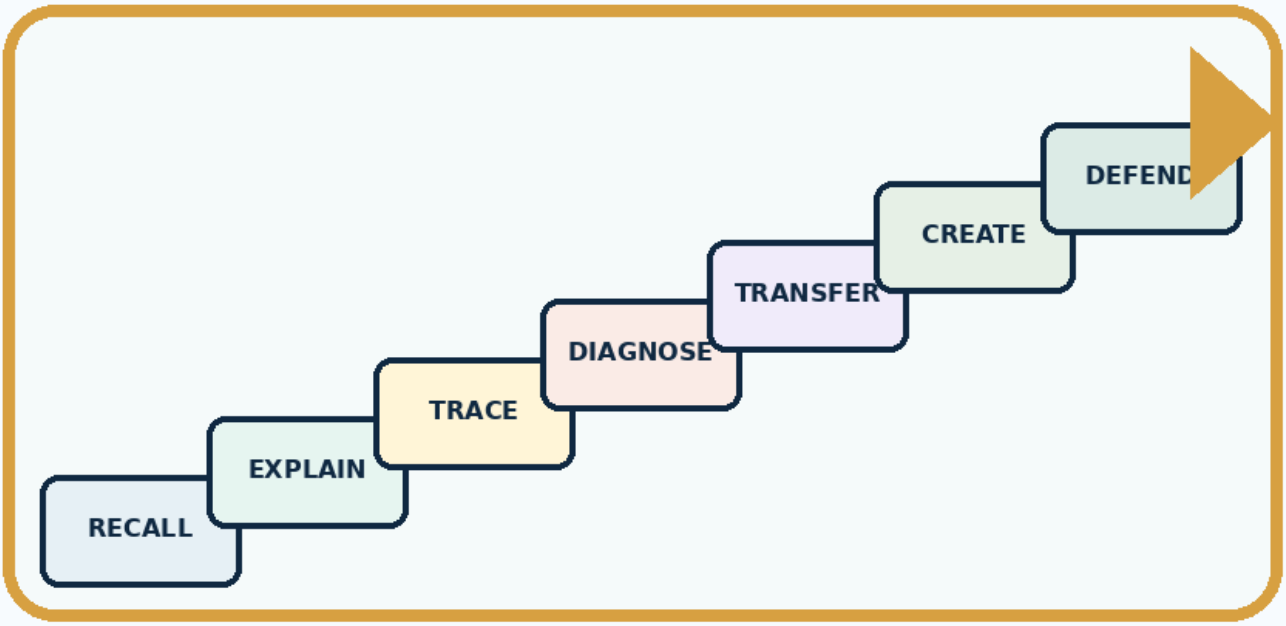
- RECALL Define **coverage**.
- EXPLAIN **Coverage**, **correctness**, and **relevance** matter more than raw quantity.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: The bigger dataset must be better.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Audit an independent paper-plane log is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Audit an independent paper-plane log is the selected independent case.



EXPLANATION CHECK Check the stated schema and source **evidence**, keep raw values, document every supported repair and report distinct assessed and unresolved records.

Independent **evidence**:

SELF-CHECK Choose the set that covers intended conditions, uses checked labels, includes relevant fields, and avoids misleading duplication. State remaining gaps rather than declaring any dataset perfect.

NEXT BRIDGE Carry today's **evidence** habit into: Missing and Unbalanced Examples.

▼ **Extra practice: Check suspicious data before changing it**

REFERENCE · SECTION 13

Extra practice: Check suspicious data before changing it

> Try the question before opening its worked explanation.

Read the supplied case and question

Ordinary pencil lengths are recorded as 12 cm, 14 cm, blank and 130 cm.

What needs checking? May 130 silently become 13?

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Investigate missing and unusually large values using the source and units. Do not invent a repair. Document verified **correction**s.

Evidence or stage	Value or result
12 cm	Plausible recorded length
14 cm	Plausible recorded length
Blank	Missing measurement
130 cm	Check source and units

▼ Earlier wording, alternatives and picture transcripts

Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Check suspicious data before changing it

Ordinary pencil lengths are recorded as 12 cm, 14 cm, blank and 130 cm.

What needs checking? May 130 silently become 13?

Investigate missing and unusually large values using the source and units. Do not invent a repair. Document verified **correction**s.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Check records against a stated format, unit and meaning before using them	Explain the deciding step, show a trace or test, and state one limit.
Distinguish an evidence-based correction from an unsupported guess	Explain the deciding step, show a trace or test, and state one limit.
Keep original values and a change log, then report what remains unresolved	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional toy-car distance table needs cleaning before use.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional toy-car distance table needs cleaning before use.
- Which labelled step matches this detail? Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**.
- 1. Clean record
- 2. **Evidence** repair
- 3. Keep unknown
- 4. Validate rows
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Clean record Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**. Cleaning is not permission to make inconvenient values plausible.

The source log supports a unit conversion and removal of a **duplicate** copy. Zero means a measured zero distance; blank means no supplied measurement.

Keep the raw table and explain every change so someone can reconstruct what happened.

Changed-case question

- Changed-case practice: The source log gives B only as bare 200 with no unit. Which conclusion is supported by the supplied facts?
- 1. B remains unresolved rather than becoming 20 cm.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: B remains unresolved rather than becoming 20 cm. Without a confirmed unit, division by 10 would be an unsupported assumption.

Repair and save your work

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone's preferences?.

Return to this week and explain what you changed.

Review Confidence Is Not **Correctness** (ai-1/18). Goal: Distinguish a confidence claim from **evidence** that a **prediction** was correct. Short practice: Was the unknown label guaranteed correct because its confidence was **80%**?.

Return to this week and explain what you changed.

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

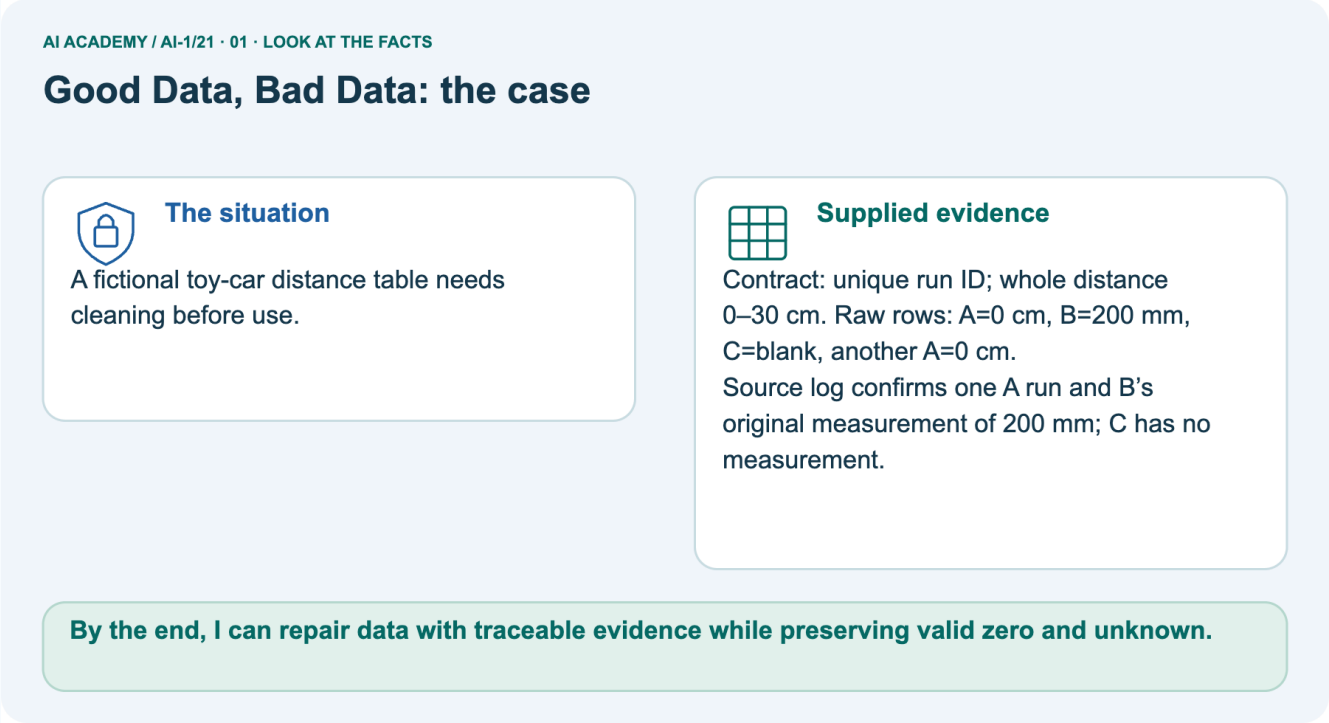
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Good Data, Bad Data: the case



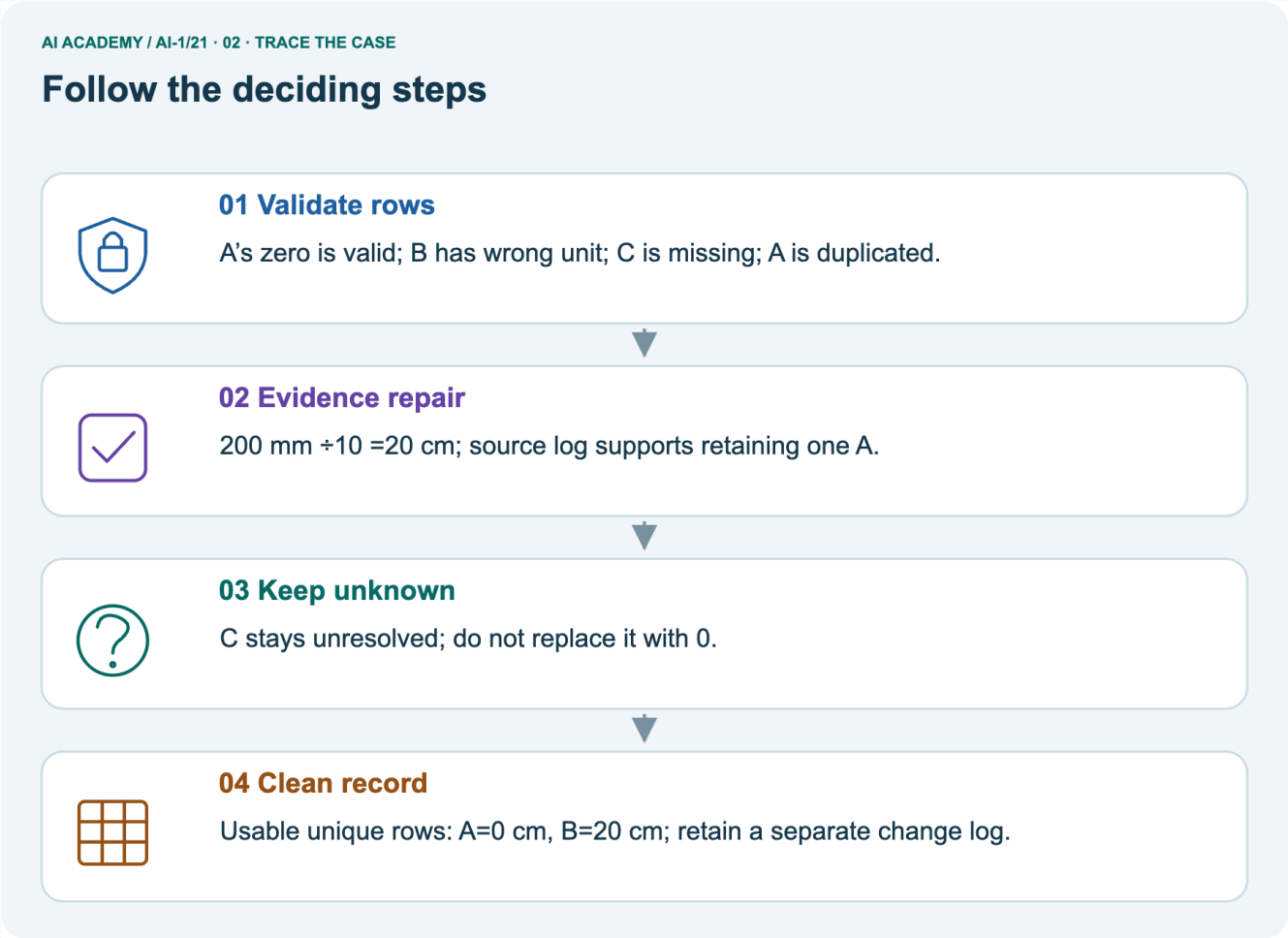
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional toy-car distance table needs cleaning before use. Supplied facts: Contract: unique run ID; whole distance 0–30 cm. Raw rows: A=0 cm, B=200 mm, C=blank, another A=0 cm.

Source log confirms one A run and B's original measurement of 200 mm; C has no measurement.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

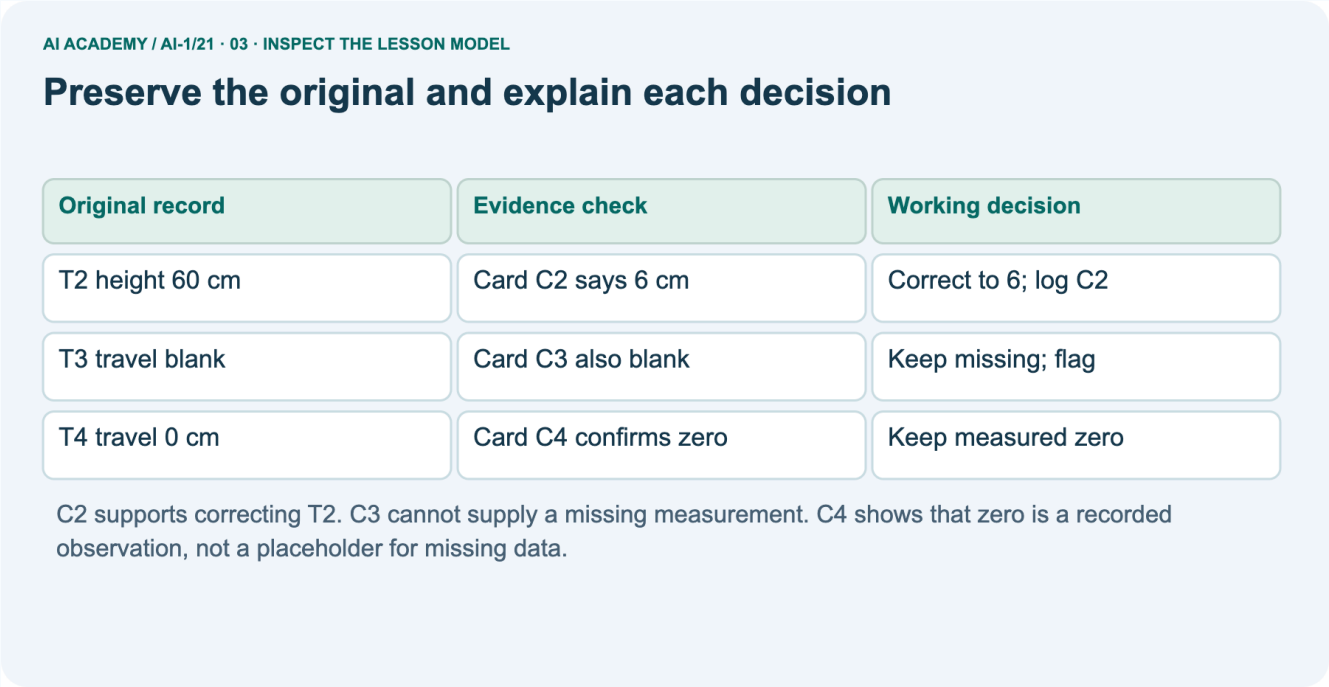


Follow the labelled picture with the matching explanation. Open the vector diagram

Validate rows: A's zero is valid; B has wrong unit; C is missing; A is **duplicate**d. → **Evidence** repair: 200 mm ÷10 =20 cm; source log supports retaining one A. → Keep unknown: C stays unresolved; do not replace it with 0. → Clean record: Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Preserve the original and explain each decision



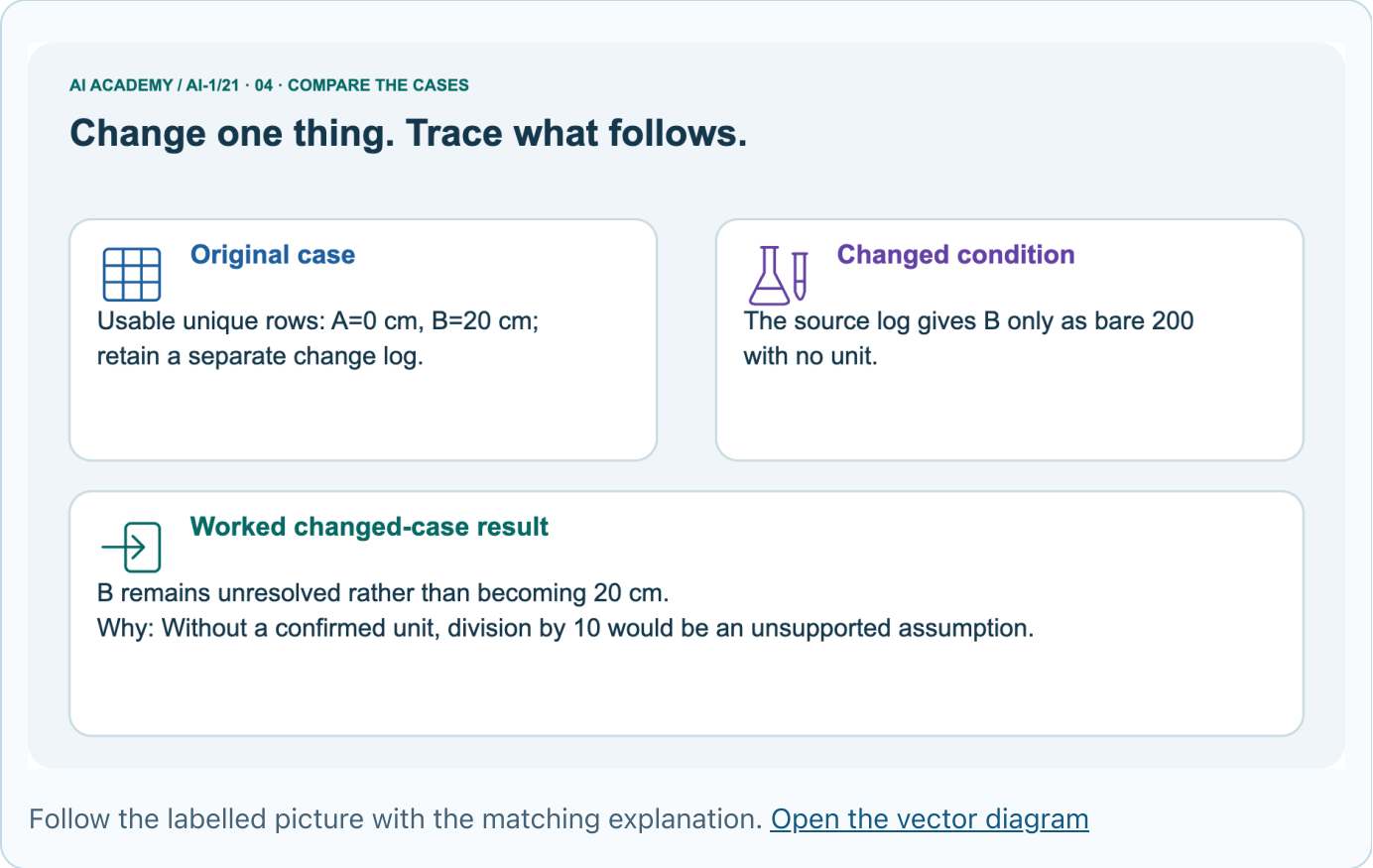
Follow the labelled picture with the matching explanation. Open the vector diagram

Preserve the original and explain each decision. T2 height 60 cm | Card C2 says 6 cm | Correct to 6; log C2. T3 travel blank | Card C3 also blank | Keep missing; flag.

T4 travel 0 cm | Card C4 confirms zero | Keep measured zero

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



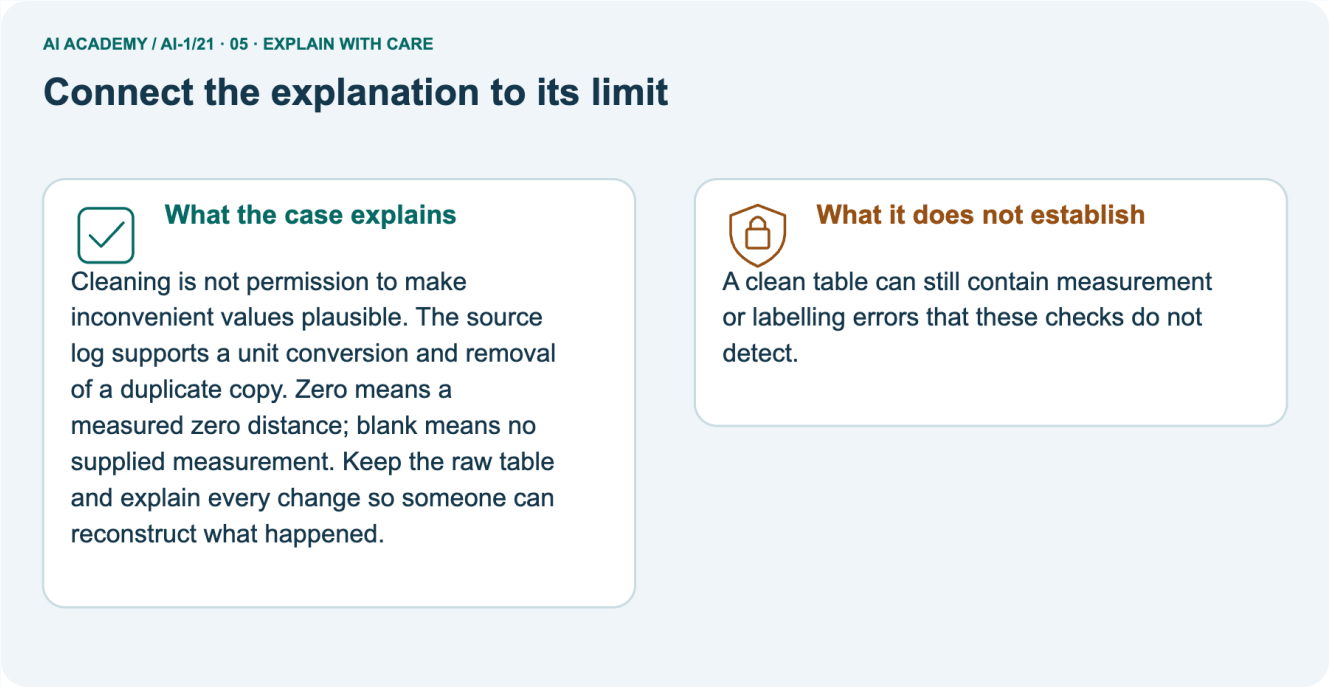
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**. Changed condition: The source log gives B only as bare 200 with no unit. Expected result: B remains unresolved rather than becoming 20 cm.

Reason: Without a confirmed unit, division by 10 would be an unsupported assumption.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



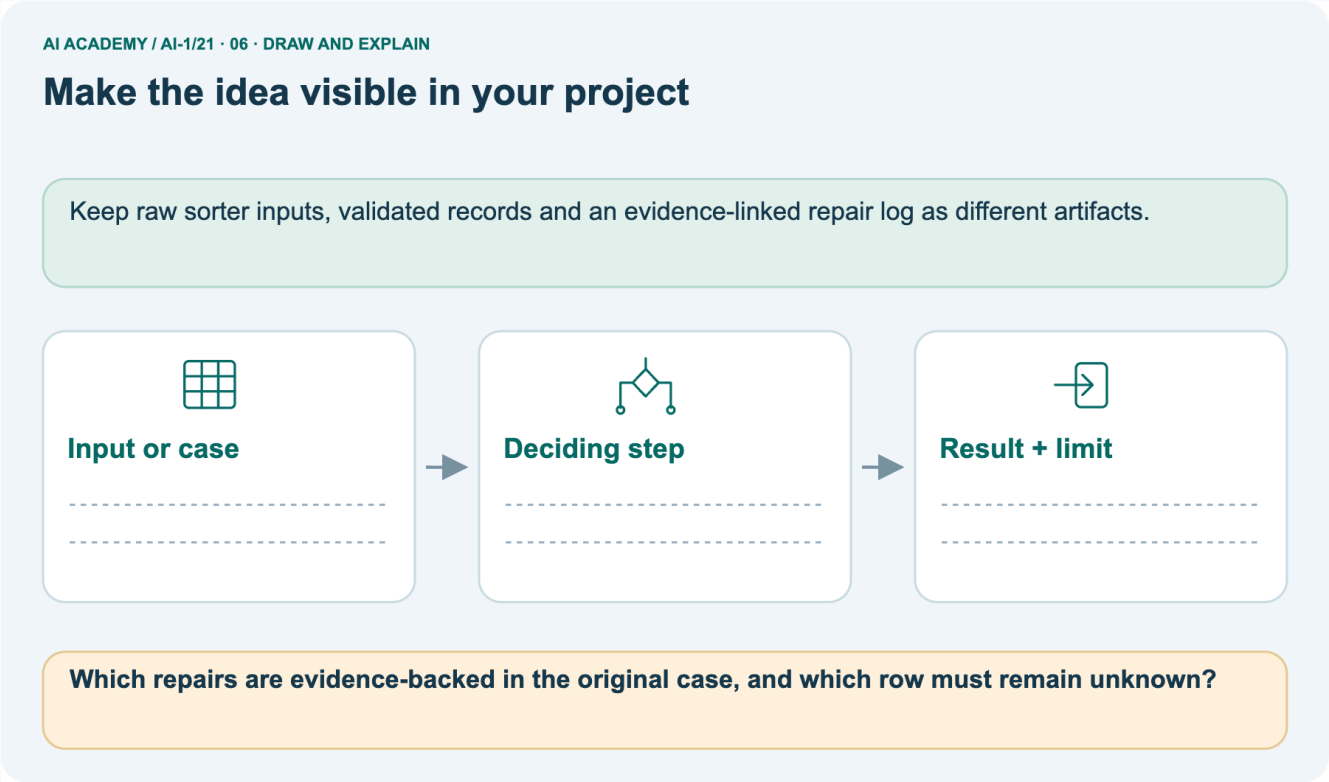
Follow the labelled picture with the matching explanation. Open the vector diagram

Cleaning is not permission to make inconvenient values plausible. The source log supports a unit conversion and removal of a **duplicate** copy. Zero means a measured zero distance; blank means no supplied measurement.

Keep the raw table and explain every change so someone can reconstruct what happened. Limit: A clean table can still contain measurement or labelling errors that these checks do not detect.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep raw sorter **input**s, validated records and an **evidence**-linked repair log as different artifacts. Draw your **input**, deciding step, result and limit.

Supported question: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Good Data, Bad Data: the case

02. Follow the deciding steps

03. Preserve the original and explain each decision

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.